MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES

COURSE NAME: SYSTEM INTEGRATION & DEPLOYMENT

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Software Requirements Specification (SRS) For the Timetable Calendar Synchronization System

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the requirements for a system that integrates the Makerere University Timetabling Management System with external calendar applications such as Google Calendar and Microsoft Outlook.

The purpose of this system is to help students easily access and manage their academic schedules by automatically synchronising timetable information with personal calendar applications. Instead of manually checking schedules, students will receive updated class information directly in their calendars.

This document is intended for developers, system designers, university IT staff, and stakeholders involved in implementing or evaluating the system.

1.2 Scope

The system will act as an integration layer between the Makerere Timetabling Management System and external calendar platforms. It will not replace the existing timetable system but will extend its functionality.

The solution introduces a Calendar Sync Service, which is a backend service responsible for retrieving timetable data, processing it, and converting it into a format that calendar applications understand.

An iCalendar (ICS) file is a standard format used to share calendar events across different systems. By generating a unique ICS feed for each student, the system allows calendar applications to subscribe and automatically receive updates.

The system will:

- Retrieve timetable data from the Makerere Timetabling Management System
- Generate a personalised ICS calendar feed for each student
- Provide a simple “Add to Calendar” feature
- Automatically update calendar events when timetable changes occur

1.3 Problem Statement (Problem Tree)

The core problem is that students miss lectures and struggle with academic time management due to inefficient access to their schedules.

This problem arises from several root causes. The Makerere Timetabling Management System currently operates in isolation, meaning it is not connected to tools that students regularly use. There is no automatic synchronisation with widely used personal calendar applications, forcing students to manually check their schedules. In addition, there are no real-time notifications when timetable changes occur, making it easy for students to miss important updates.

Because of these issues, several negative effects occur. Students forget classes, fail to notice timetable clashes, and struggle to properly plan their academic activities. This leads to increased stress, poor time management, and ultimately reduced academic performance and learning effectiveness.

1.4 Proposed Solution

To solve this problem, an integration solution is proposed that connects the Makerere Timetabling Management System with external calendar applications using the iCalendar (ICS) format.

A Calendar Sync Service will act as an intermediary system. This means it will sit between the timetable system and external calendar platforms. It will retrieve student timetable data, process it, and generate a personalised calendar feed for each student.

Students will be able to subscribe to their timetable using a simple "Add to Calendar" feature. Once subscribed, their calendar application will automatically receive updates whenever changes occur in the timetable system. This includes changes such as lecture time adjustments, venue changes, or course updates.

This solution removes the need for manual checking and ensures that students always have up-to-date schedule information.

1.5 Definitions and Key Terms

An API (Application Programming Interface) is a way for different software systems to communicate with each other.

An ICS (iCalendar) file is a standard file format used to store and share calendar data such as events, dates, and times.

A Calendar Feed is a URL link that allows a calendar application to subscribe and receive updates automatically.

1.6 References

- Internet Engineering Task Force – iCalendar Specification (RFC 5545)
- Uganda Fourth National Development Plan
- United Nations Sustainable Development Goals (Goal 4: Quality Education)

2. Overall Description

2.1 System Perspective

The system operates as an extension of the existing Makerere Timetabling Management System. The Calendar Sync Service retrieves timetable data from this system and transforms it into a format usable by external calendar applications. These applications then display the schedule to students in a familiar interface.

2.2 System Functions

The system performs several key functions in a continuous and automated way. It retrieves timetable data, processes it into structured calendar events, and publishes it as an ICS feed. When any change occurs in the timetable system, the service updates the feed so that subscribed calendars reflect the new information.

It also provides a simple way for students to connect their calendars through a one-click subscription feature.

2.3 Stakeholders

The main stakeholders include students, who are the primary users of the system, and university IT staff who will maintain and support the integration. Timetable administrators are responsible for managing schedule data, while external calendar providers act as platforms that display the synchronised information.

The project team, consisting of Kau Vincent James, Okwir Moses, and Nkulikiyumukiza Jean D'amour, is responsible for designing and implementing the system.

2.4 Operating Environment

The system will run in a web-based environment and will be hosted on the cloud infrastructure. It will communicate with external systems over the internet using secure protocols.

It must be compatible with widely used calendar applications and modern web browsers such as Google Calendar and Microsoft Outlook.

2.5 Constraints and Assumptions

The system depends on the availability and accuracy of the Makerere Timetabling Management System. It assumes that timetable data can be accessed through a database or API.

It also assumes that students have access to internet-enabled devices and use calendar applications that support ICS subscriptions.

3. Functional Requirements (FR)

FR01: The system must be able to generate a unique calendar feed for each student. This feed will contain all relevant timetable information, including course names, lecturers, times, and venues. Each event must be clearly structured so that it appears correctly in calendar applications.

FR02: The system must ensure that timetable updates are automatically reflected in the student's calendar. This means that when a class is rescheduled or a venue changes, the updated information is pushed to the calendar without requiring any action from the student.

FR03: A simple and user-friendly "Add to Calendar" feature must be provided. This feature allows students to subscribe to their timetable using a single action. Once subscribed, no further setup should be required.

FR04: Each student must have a unique and secure calendar feed link. This ensures that only the intended user can access their schedule.

4. Non-Functional Requirements (NFR)

NFR01: Security: Calendar feeds should be protected using long, unique tokens that are difficult to guess. All communication between systems must be encrypted using HTTPS.

NFR02: Reliability: Updates from the timetable system should be reflected in external calendars within a reasonable time, depending on how often the calendar application refreshes the feed.

NFR03: Compatibility: The system must follow the iCalendar (RFC 5545) standard to ensure it works with different calendar platforms without errors.

NFR04: Scalability: It should be able to handle a large number of students without performance issues. Techniques such as caching may be used to improve efficiency.

NFR05: Usability: The system should be simple to use, requiring minimal effort from students. Once a student subscribes to their calendar, the system should work automatically without additional configuration.

5. Data Requirements

The system will use several types of data. This includes student identification details like student and registration number, course information, enrolment records, and timetable schedules.

It will also store calendar feed tokens, which are unique identifiers used to secure access to each student's calendar feed.

Calendar event data will be generated dynamically from timetable information. This includes event titles, start and end times, locations, and descriptions.

6. External Interfaces

The system will connect directly to the Makerere Timetabling Management System. This connection is essential for retrieving accurate timetable data.

It will also interact with external calendar systems through ICS feeds and optionally through APIs such as Google Calendar API or Microsoft Graph API.

7. Alignment with Development Frameworks

This system supports national and global development goals. It aligns with Uganda's digital transformation strategy under the Uganda Fourth National Development Plan by promoting the use of technology to improve service delivery in education.

It also contributes to the United Nations Sustainable Development Goal 4, which focuses on improving access to quality education. By helping students better manage their schedules, the system enhances learning effectiveness and academic success.

8. Acceptance Criteria

The system will be considered successful if students are able to subscribe to their timetable and view all their classes correctly in their calendar applications. Any updates made in the timetable system must be reflected automatically in the calendar.

The "Add to Calendar" feature must work reliably, and calendar feeds must be secure and accessible only to the intended student.

The system must also demonstrate compatibility with at least major calendar platforms such as Google Calendar and Microsoft Outlook.